

Calibre D2S R&D C++ Project Report

By Andrew Morgan

- **Platform and Compiler details:**

- Platform used is: Windows OS
- Built and Compiled using Visual Studio 2019 with the MSVC (Microsoft Visual C++) compiler, version 19.29, supporting C++17.

- **Diagrams and Visual Representations:**

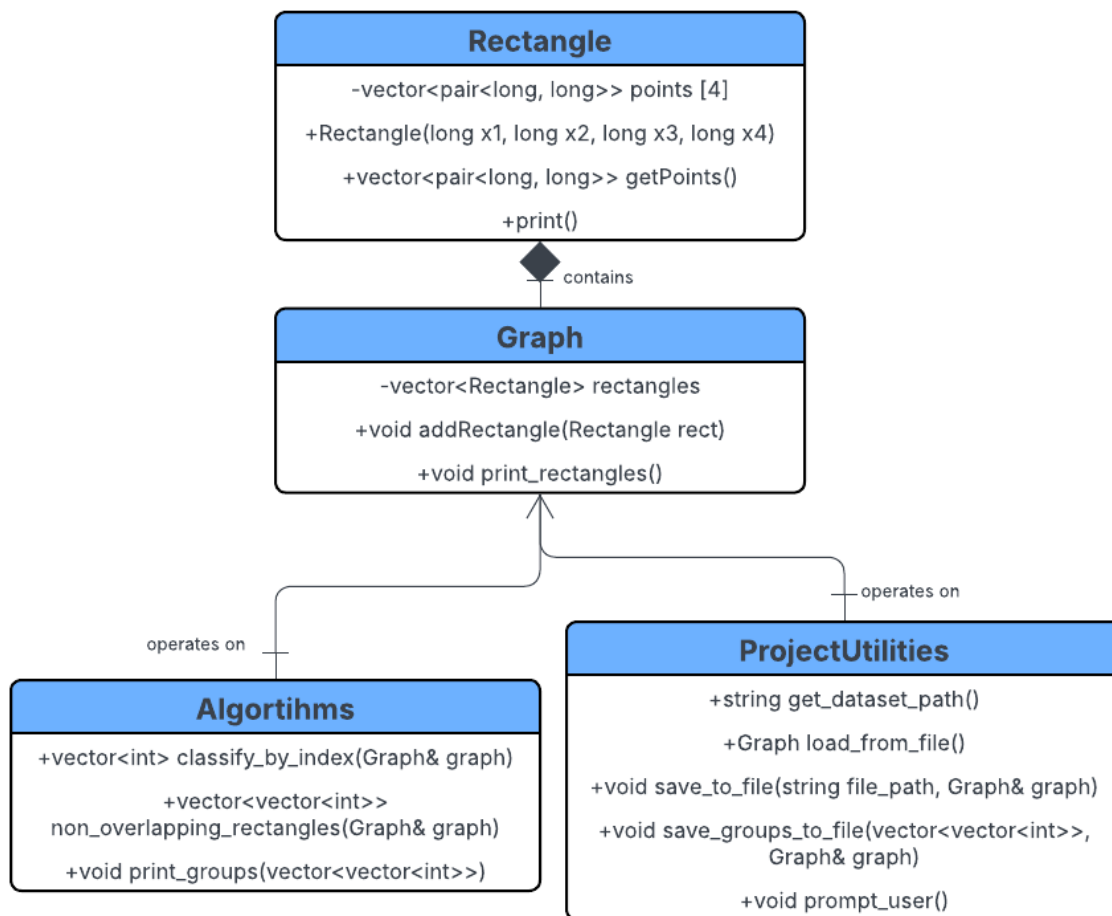


Figure 1: Project UML Diagram

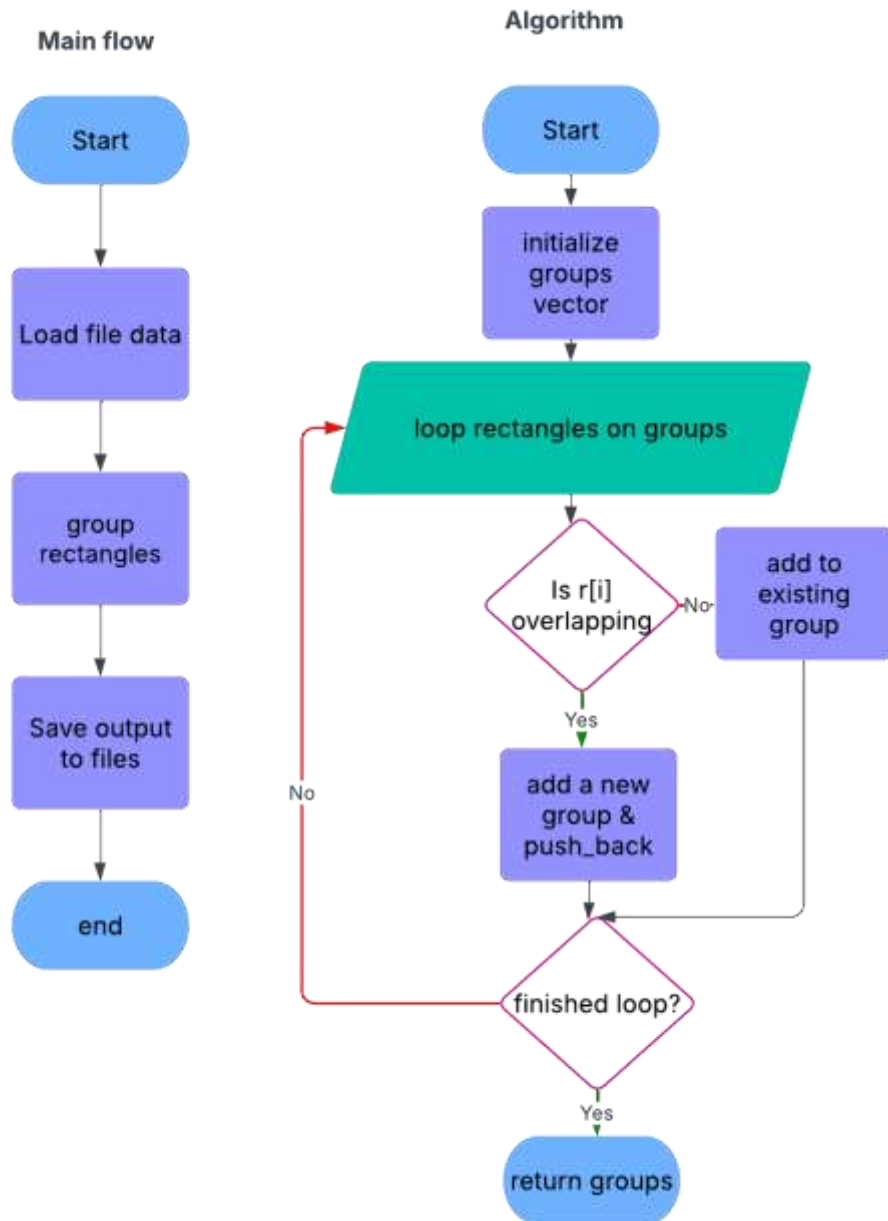


Figure 2: Project Flowchart

- Algorithm Pseudo Code:

```

FUNCTION non_overlapping_rectangles(graph):
    groups ← NULL vector of vectors
    rectangles ← dataset rectangles
    sort rectangles by x1
    FOR each rectangle index i in rectangles:
        placed ← FALSE
        FOR each group in groups:
            overlaps ← FALSE
            FOR each index in group:
                IF NOT (this rectangles[i] is completely left OR
                    is completely right OR
                    is completely below OR
                    is completely above rectangles[index] of group[ind]):
                    overlaps ← TRUE
                    BREAK
            IF NOT overlaps:
                ADD i to group
                placed ← TRUE
                BREAK
        IF NOT placed: // If rectangle i does not fit in any existing group, create a new group
            ADD [i] as a new group in groups
    RETURN groups
  
```

- Diagrams and Visual Representations:

Data set	Number of inputs	Number of output groups	Runtime (sec)	Memory Usage (MB)
data_set_1.txt	5	2	0.002151	0.007813
data_set_2.txt	7	2	0.001273	0
data_set_3.txt	20	5	0.002799	0.011719
data_set_4.txt	39	4	0.003158	0
data_set_5.txt	77	3	0.005035	0.015625
data_set_6.txt	136	5	0.009196	0.019531
data_set_7.txt	216	5	0.013341	0.035156
data_set_8.txt	460	5	0.02864	0.113281
data_set_9.txt	741	12	0.046866	0.144531
data_set_10.txt	981	6	0.077383	0.25
data_set_11.txt	5793	7	1.28866	1.01172
data_set_12.txt	6775	7	1.67023	1.09375
data_set_13.txt	7538	7	2.03496	1.00781
data_set_14.txt	8774	7	2.69093	1.66016
data_set_15.txt	9188	6	2.91699	1.09766
data_set_16.txt	N/A	N/A	N/A	N/A

- **Testing Methodology:**

- Manually designed testcases for possible edge cases and wrong entries.

Example testcases:

I. Basic:

- Non-Overlapping rectangles.
- Overlapped rectangles.

II. Corner Cases:

- Touching edges.
- Touching corners.
- Sorted and Non-Sorted rectangles.

III. Errors and Invalid data:

- $P1 = P2, X1 = X2, Y1 = Y2$.
- Non numeric values.
- More than specified range ($> LONG_MAX, < LONG_MIN$).
- Missing data (less than 4 values per line).
- Extra data (more than 4 values per line).
- Empty file.

- **General Notes:**

- Huge data_set_16 causes a runtime error while reading rectangles.
- Access specifiers are mostly public for simplicity.
- Code style used is Hybrid; camel case for classes and snake case for variables and functions.
- There are multiple debugging and other functions existing in the project, main functions are the ones in the UML and Flowchart, also an uncompleted algorithm.

- Screenshots:

Name	Date modified	Type	Size
Debug	16-Feb-25 9:28 PM	File folder	
input	16-Feb-25 9:18 PM	File folder	
internship_project_datasets	13-Feb-25 11:50 AM	File folder	
output	16-Feb-25 4:04 PM	File folder	
Algorithms.cpp	16-Feb-25 9:43 PM	C++ Source	3 KB
Algorithms.h	16-Feb-25 9:43 PM	C/C++ Header	1 KB
Graph.cpp	16-Feb-25 9:37 PM	C++ Source	1 KB
Graph.h	16-Feb-25 9:37 PM	C/C++ Header	1 KB
Header.h	10-Jun-21 7:00 AM	C/C++ Header	1 KB
ProjectUtilities.cpp	16-Feb-25 9:50 PM	C++ Source	13 KB
ProjectUtilities.h	16-Feb-25 9:50 PM	C/C++ Header	1 KB
Rectangle.cpp	16-Feb-25 9:37 PM	C++ Source	1 KB
Rectangle.h	16-Feb-25 9:36 PM	C/C++ Header	1 KB
Rectangles.vcxproj	15-Feb-25 10:51 PM	VC++ Project	9 KB
Rectangles.vcxproj.filters	15-Feb-25 10:51 PM	VC++ Project File...	2 KB
Rectangles.vcxproj.user	13-Feb-25 11:45 AM	USER File	1 KB
Source.cpp	16-Feb-25 9:50 PM	C++ Source	1 KB

Screenshot 2: Project Hierarchy

```
// main.cpp
#include "ProjectUtilities.h"

int main() {
    ProjectUtilities::process_input_method();
    return 0;
}
```

C:\Users\Andrew\source\repos\Rectangles\D... — □ ×

Select input method:
 1. Process a single dataset
 2. Process all datasets (small, medium, large)
 3. Process all files in directory
 Enter choice:

Screenshot 2: main.cpp and console interface

Name	Date modified
large	16-Feb-25 1:50 PM
medium	16-Feb-25 1:50 PM
random	16-Feb-25 3:23 PM
small	16-Feb-25 1:50 PM
all_datasets_results.csv	16-Feb-25 10:03 PM

Screenshot 3: output directory

```
Microsoft Visual Studio Debug Console

Dataset: data_set_4 | Runtime: 0.0029566s | Memory Usage: 0.0078125 MB
Processing small -> data_set_5
Dataset: data_set_5 | Runtime: 0.0062546s | Memory Usage: 0.0117188 MB
Processing medium -> data_set_6
Dataset: data_set_6 | Runtime: 0.0104062s | Memory Usage: 0.03125 MB
Processing medium -> data_set_7
Dataset: data_set_7 | Runtime: 0.0178707s | Memory Usage: 0.0742188 MB
Processing medium -> data_set_8
Dataset: data_set_8 | Runtime: 0.0412626s | Memory Usage: 0.113281 MB
Processing medium -> data_set_9
Dataset: data_set_9 | Runtime: 0.0842277s | Memory Usage: 0.253906 MB
Processing medium -> data_set_10
Dataset: data_set_10 | Runtime: 0.117558s | Memory Usage: 0.246094 MB
Processing large -> data_set_11
Dataset: data_set_11 | Runtime: 2.58737s | Memory Usage: 1.26953 MB
Processing large -> data_set_12
Dataset: data_set_12 | Runtime: 3.45735s | Memory Usage: 1.13281 MB
Processing large -> data_set_13
Dataset: data_set_13 | Runtime: 4.22803s | Memory Usage: 0.925781 MB
Processing large -> data_set_14
Dataset: data_set_14 | Runtime: 5.68708s | Memory Usage: 1.27344 MB
Processing large -> data_set_15
Dataset: data_set_15 | Runtime: 6.19461s | Memory Usage: 1.73438 MB
All datasets processed successfully!

C:\Users\Andrew\source\repos\Rectangles\Debug\Rectangles.exe (process 11248) exited with code 0.
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

Screenshot 4: running all datasets

```
Microsoft Visual Studio Debug Console

Warning: Values exceed long range. Skipping this rectangle.
Error: No valid rectangles found in the file. The file may be empty or all rectangles were invalid.
Processing file: input\extra_value.txt
Warning: Invalid rectangle: x1 must be different and smaller from x2 and y1 must be different and smaller from y2. Skipping this rectangle.
Error: No valid rectangles found in the file. The file may be empty or all rectangles were invalid.
Processing file: input\less_value.txt
Warning: Invalid input format. Skipping this line: 1 2 3
Error: No valid rectangles found in the file. The file may be empty or all rectangles were invalid.
Processing file: input\long_max_greater.txt
Warning: Invalid input format. Skipping this line: 18446744073709551610 18446744073709551610 18446744073709551610 18446744073709551610
Error: No valid rectangles found in the file. The file may be empty or all rectangles were invalid.
Processing file: input\long_min_lower.txt
Warning: Invalid input format. Skipping this line: -18446744073709551610 -18446744073709551610 -18446744073709551610 -18446744073709551610
Error: No valid rectangles found in the file. The file may be empty or all rectangles were invalid.
Processing file: input\non_numerical_values.txt
Warning: Invalid input format. Skipping this line: -9 p 2 0
Error: No valid rectangles found in the file. The file may be empty or all rectangles were invalid.
Processing file: input\sorted_random - Copy.txt
Processing file: input\sorted_random.txt
Processing file: input\straight_lines.txt
Warning: Invalid rectangle: x1 must be different and smaller from x2 and y1 must be different and smaller from y2. Skipping this rectangle.
Warning: Invalid rectangle: x1 must be different and smaller from x2 and y1 must be different and smaller from y2. Skipping this rectangle.
Error: No valid rectangles found in the file. The file may be empty or all rectangles were invalid.
Processing file: input\unsorted_random - Copy.txt
Processing file: input\unsorted_random.txt

C:\Users\Andrew\source\repos\Rectangles\Debug\Rectangles.exe (process 6300) exited with code 0.
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

Screenshot 5: running custom datasets from directory input